



OneATS

User Manual

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TAKAOKA TOKO CO., LTD.



OneATS user manual

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Definitions, symbols, abbreviations

Symbol	Description
CIM	Common Information Model
DMS	Distributed Management System
DCOM	Distributed Component Object Model
FEP	Front End Processing
HMI	Human Machine Interface
I/O	Input/Output
IED	Intelligent Electronic Device
LAN	Local Area Network
OPC UA	OPC Unified Architecture
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SSL	Secure Socker Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
XML	Extensible Markup Language

Text color:

Comment

Notes

Warnings

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A.	System overview
B.	Database
C.	OneATS DataEditor – Project Management
D.	OneATS FEP
E.	OneATS DataServer
F.	OneATS Grid Studio
G.	OneATS SmartHIS
H.	OneMasterTool



OneATS User Manual

- OneMasterTool

14/Jan/2025

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Software Version	Date of Issue	Change Description
V4.2.0.6	26/Apr/2024	
V4.2.0.7	24/Jun/2024	Upgrade the internal configuration of the software
V4.2.0.9	12/Aug/2024	Configure HIS on OneMasterTool - added Ignored Qualities parameter

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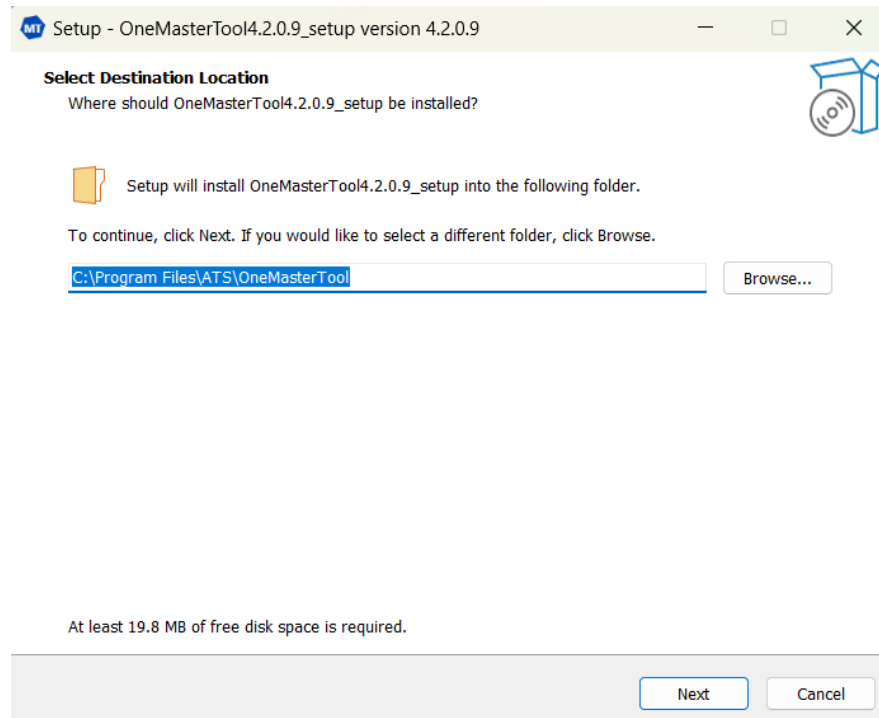
H. OneMasterTool

I. OneMasterTool Overview

OneMasterTool is a tool that supports configuring and activating licenses for software modules in OneATS.

II. Install the app

Run the installation file OneMasterTool.exe. Select **Next** to install:



OneMasterTool Shortcut on Desktop:

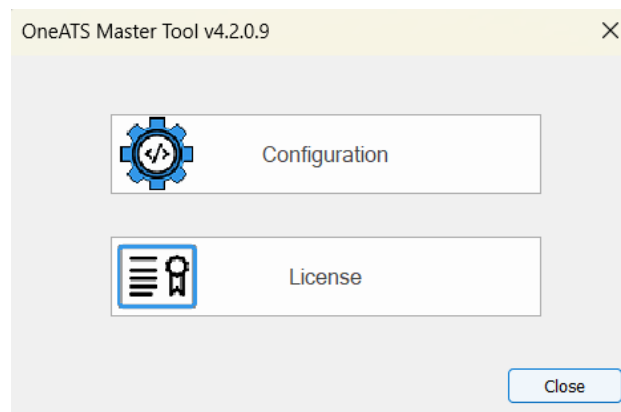


Figure H-1. OneMasterTool Shortcut on Desktop

III. Configuring Software Modules on OneMasterTool

Run the OneMasterTool application, the application window is divided into two main parts: Configuration and License. In which:

- **Configuration:** Tool for configuring software modules in OneATS.
- **License:** Tool to activate licenses of software modules in OneATS.

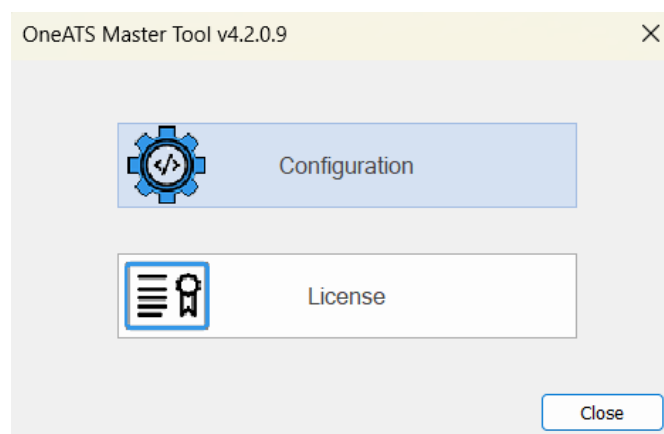


1. Create a Profile on OneMasterTool

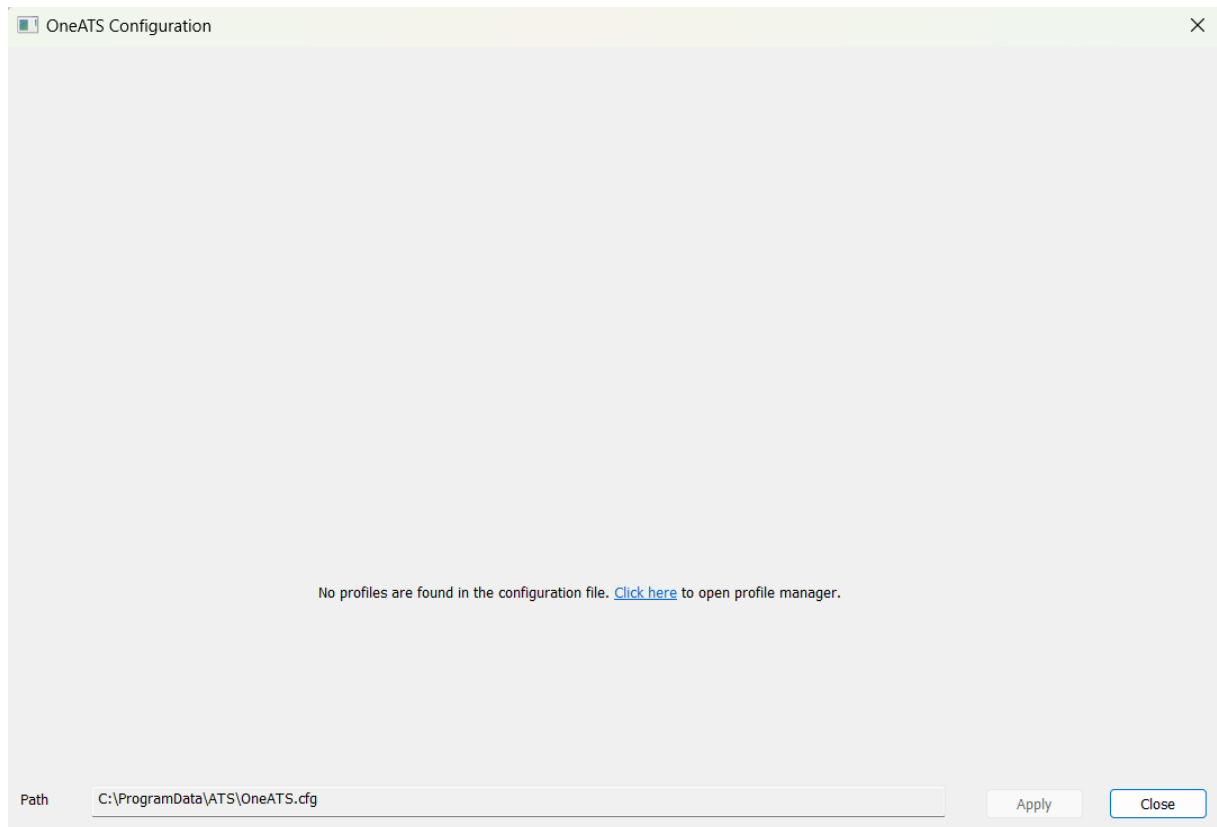
OneMasterTool application supports configuring multiple profiles in a configuration file and the profile option is enabled. When a profile is selected as enabled, the configuration information of this profile will be applied to the software modules.

Run the OneMasterTool application with **Administrator permissions**, select

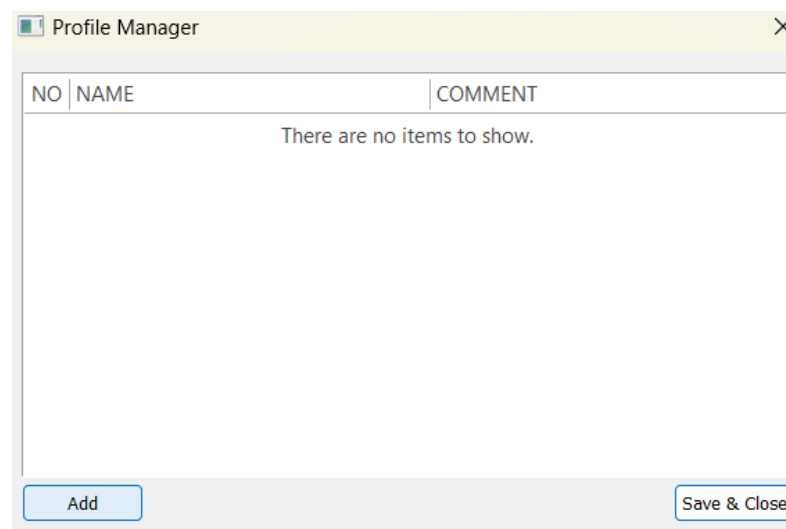
Configuration:



Select **Click here** to open the **Profile Manager** window:



Select **Add** to add a Profile:



Enter **Name**, **Comment**, select the profile that has the function of configuring which software modules in **Select Module**, select **Ok**:

Profile Manager - New

Name

Comment

Copy from

Select Module

- ☒ Database
- ☒ Data Server
- ☒ FEP
- ☒ HIS

OK Cancel

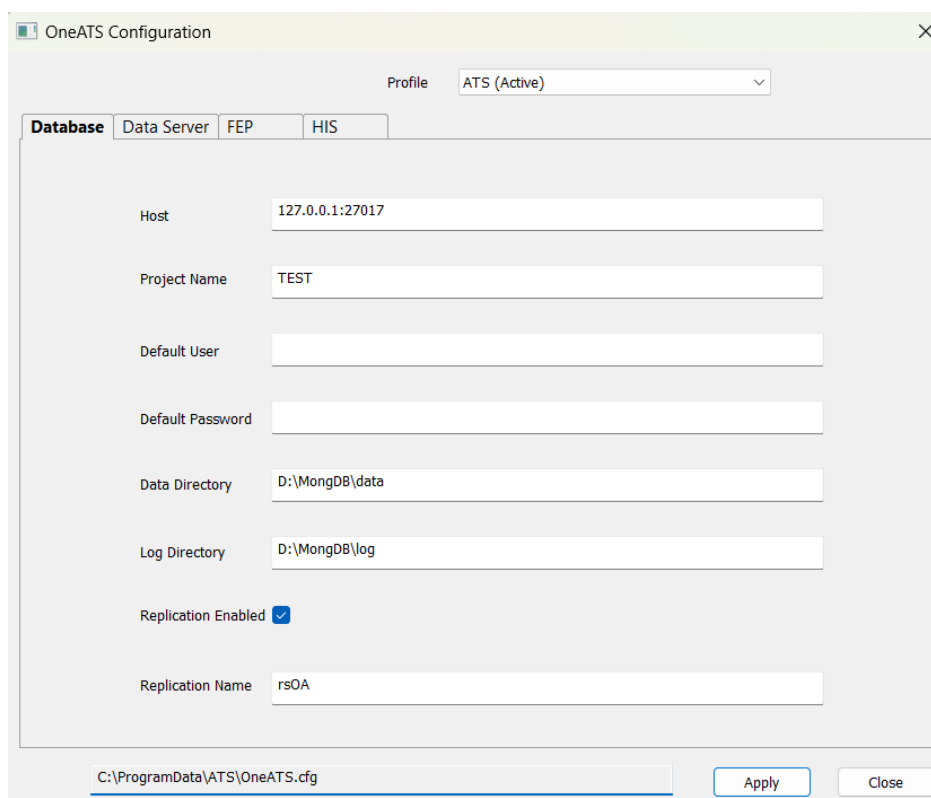
Select **Save & Close** to save and close the Profile Manager window.

Profile Manager

NO	NAME	COMMENT
1	ATS	ATS

Add Save & Close

OneMasterTool Interface:



The OneATS Configuration window shows the following settings:

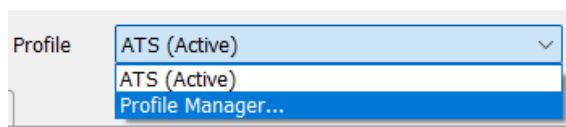
- Profile: ATS (Active)
- Database: Selected tab
- Data Server: Empty
- FEP: Empty
- HIS: Empty
- Host: 127.0.0.1:27017
- Project Name: TEST
- Default User: Empty
- Default Password: Empty
- Data Directory: D:\MongDB\data
- Log Directory: D:\MongDB\log
- Replication Enabled: ☒
- Replication Name: rsOA

File path: C:\ProgramData\ATS\OneATS.cfg

Buttons: Apply, Close

Figure H-2. OneMasterTool Interface

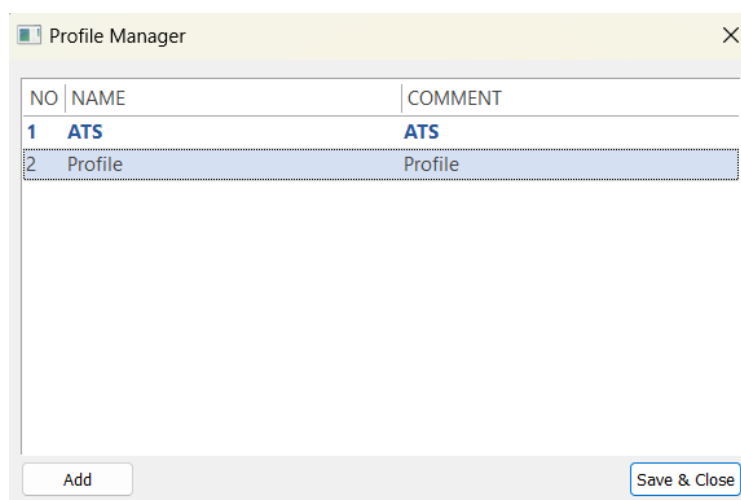
To manage Profiles, select **Profile >> Profile Manager ...**



Profile dropdown menu showing:

- ATS (Active)
- ATS (Active)
- Profile Manager...

The Profile Manager interface appears, users can add, edit, delete, and Set Active for profile.

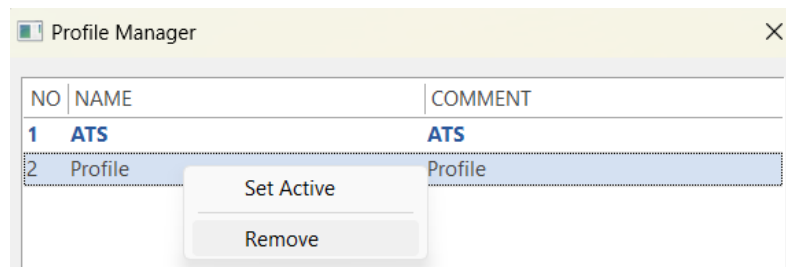


NO	NAME	COMMENT
1	ATS	ATS
2	Profile	Profile

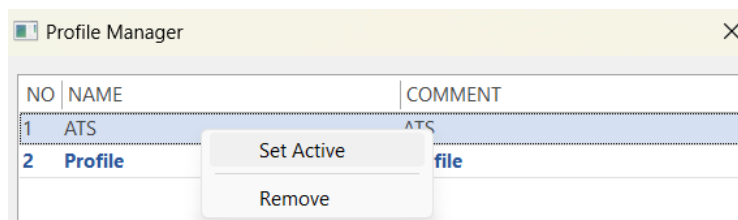
Buttons: Add, Save & Close

- Add new profile: On the Profile Manager interface, select **Add** to add a profile, enter **Name**, **Comment**, **Copy from** (get configuration information from an existing profile) and select software modules in **Select Module**.

- Edit profile: On the Profile Manager interface, Double click on the name or description of the profile, enter the information needed to be changed.
- Delete profile: On the Profile Manager interface, Right click on the profile you want to delete and select **Remove**.



- Configure Set Active for profile: On the Profile Manager interface, Right click on the profile and select **Set Active**.



After making changes, select **Save & Close** to save and close the Profile Manager window.

2. Configuring the Database on OneMasterTool

Run the OneMasterTool application, select **Configuration**, select the **Database** tab:

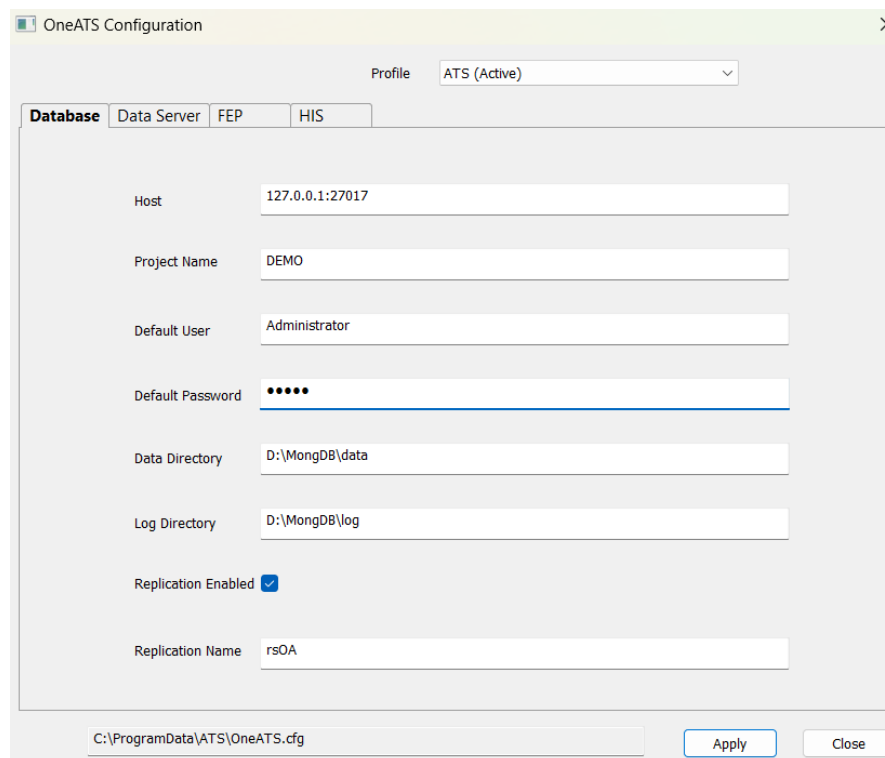


Figure H-3. Database Configuration Interface on OneMasterTool

In which:

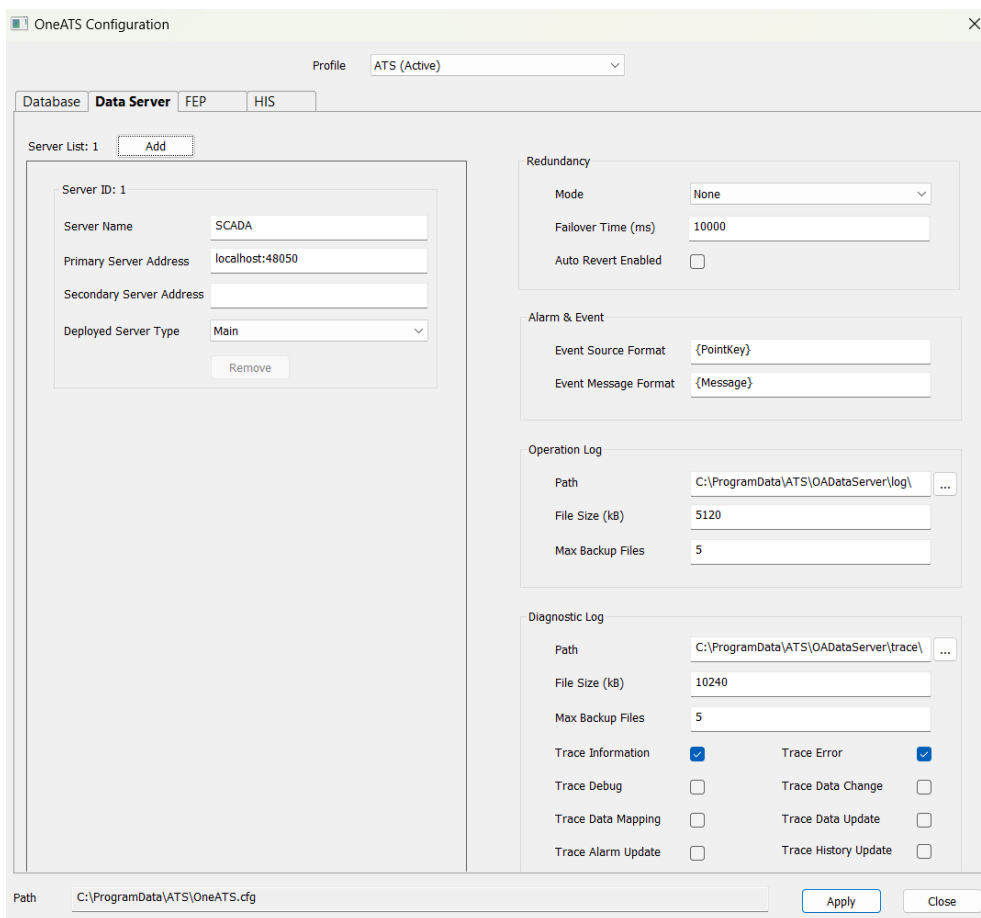
- **Host:** The address and port of the MongoDB Server, in case of deploying multiple MongoDB Servers on multiple machines, these values are separated from each other by a ";" sign.
- **Project Name:** The name of the project.
- **Data Directory:** The path to the address where the data is stored.
- **Log Directory:** The path to the log storage address.
- **Replication Enabled:** Enable/Disable Replication.
- **Replication Name:** The name of the Replication, used for the Enable Replication case.

Select **Apply** to save the configuration file.

When you select **Apply**, the database configuration will be overwritten in the `mongod.cfg` file

3. Configure DataServer on OneMasterTool

Run the OneMasterTool application, select **Configuration**, select the **DataServer** tab:



The screenshot displays the 'OneATS Configuration' window with the 'Data Server' tab selected. The 'Profile' dropdown is set to 'ATS (Active)'. The 'Database' tab is also visible. The 'Server List' section shows one server with ID 1, named 'SCADA', with a primary address of 'localhost:48050' and a deployed type of 'Main'. The 'Redundancy' section has 'Mode' set to 'None', 'Failover Time (ms)' set to '10000', and 'Auto Revert Enabled' unchecked. The 'Alarm & Event' section has 'Event Source Format' set to '{PointKey}' and 'Event Message Format' set to '{Message}'. The 'Operation Log' section has 'Path' set to 'C:\ProgramData\ATS\OADataServer\log\', 'File Size (kB)' set to '5120', and 'Max Backup Files' set to '5'. The 'Diagnostic Log' section has 'Path' set to 'C:\ProgramData\ATS\OADataServer\trace\', 'File Size (kB)' set to '10240', and 'Max Backup Files' set to '5'. The 'Trace Information' section has 'Trace Information' and 'Trace Error' checked, while 'Trace Debug', 'Trace Data Change', 'Trace Data Mapping', 'Trace Data Update', 'Trace Alarm Update', and 'Trace History Update' are unchecked. The 'Path' field at the bottom shows 'C:\ProgramData\ATS\OneATS.cfg'. The 'Apply' and 'Close' buttons are at the bottom right.

Figure H-4. DataServer Configuration Interface on OneMasterTool

In which:

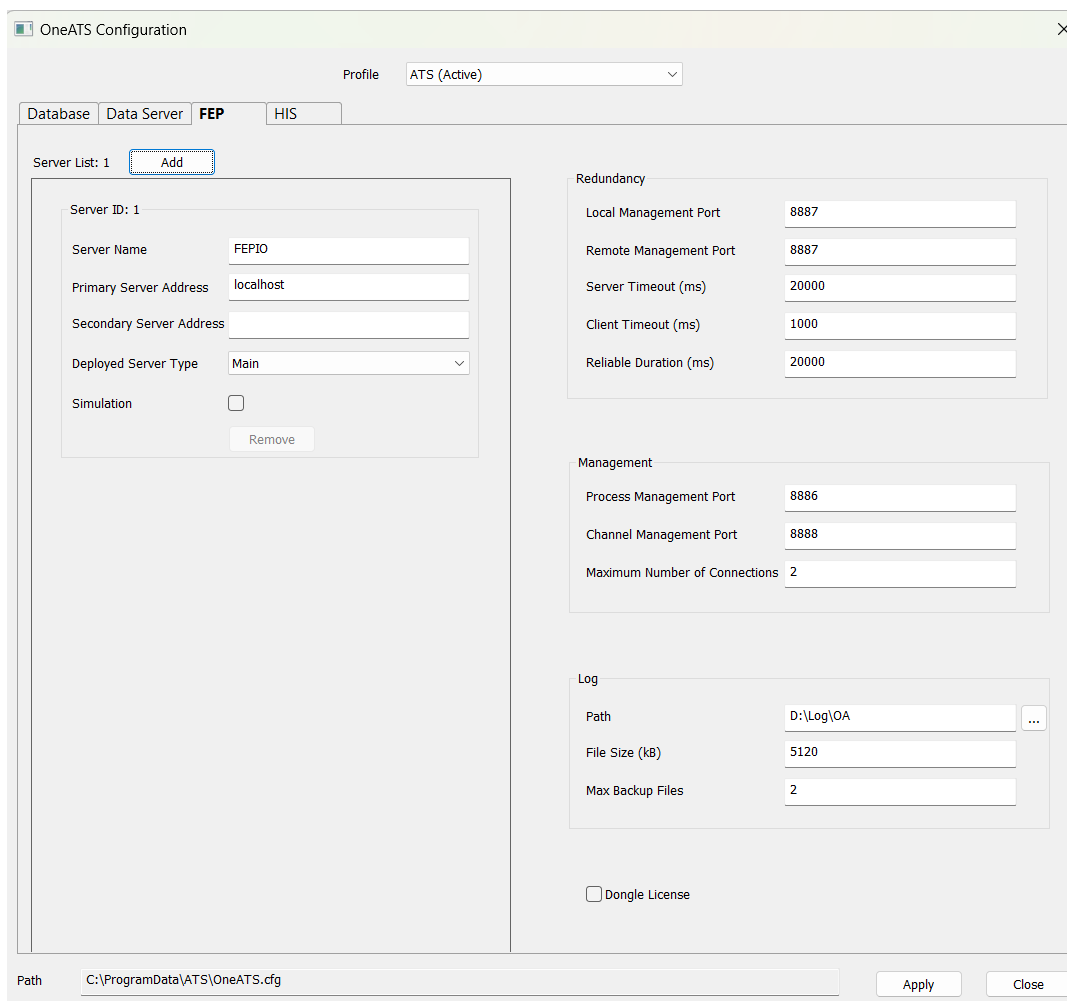
- **Server Name:** The name of the server.
- **Primary Server Address:** The server address and data port of the primary server.
- **Redundancy Server Address:** The address and data port of the redundancy server.
- **Deployed Server Type:** The option is server main/backup/none.
- **Redundancy:**
 - Mode modes: None, Warm, Hot.
 - **Failover Time:** Failover time.
 - **Auto-Revert Enabled:** Determines what happens when the primary server comes back online after the failover takes place. When checked, the primary server will resume the running mode when it comes back online, the backup server will switch to the standby mode.
- **Alarms & Events:**
 - **EventSourceFormat:** The Source format of the Event, Alarm.
 - **EventMessageFormat:** The message format of the Event, Alarm.
- **Operation Log:**
 - **Path:** The location where the logs are stored.
 - **File Size (kB):** The size of the log file.
 - **Max Backup Files:** The number of log files that are backed up when the log file reaches the maximum size.

Select **Apply** to save the configuration file.

The Server parameter configuration matches the **SCADA System** configuration on OneATS DataEditor.

4. Configure FEP on OneMasterTool

Run the OneMasterTool application, select **Configuration**, select the **FEP** tab:



The screenshot shows the 'OneATS Configuration' window with the 'FEP' tab selected. The 'Profile' dropdown is set to 'ATS (Active)'. The 'Server List' section on the left contains one entry, 'Server ID: 1', with fields for 'Server Name' (FEP10), 'Primary Server Address' (localhost), 'Secondary Server Address' (empty), 'Deployed Server Type' (Main), and 'Simulation' (unchecked). The 'Redundancy' section on the right includes fields for 'Local Management Port' (8887), 'Remote Management Port' (8887), 'Server Timeout (ms)' (20000), 'Client Timeout (ms)' (1000), and 'Reliable Duration (ms)' (20000). The 'Management' section includes 'Process Management Port' (8886), 'Channel Management Port' (8888), and 'Maximum Number of Connections' (2). The 'Log' section includes 'Path' (D:\Log\OA), 'File Size (kB)' (5120), and 'Max Backup Files' (2). At the bottom, there is a 'Dongle License' checkbox and a 'Path' field showing 'C:\ProgramData\ATS\OneATS.cfg'. 'Apply' and 'Close' buttons are at the bottom right.

Figure H-5. FEP Configuration Interface on OneMasterTool

In which:

- **Server List:** List of Fep Servers with attributes:
 - **Server ID:** The ID of the server.
 - **Server Name:** The name of the server.
 - **Primary Server Address:** The address of the machine where FEP main is installed
 - **Secondary Server Address:** The address of the machine where FEP backup is installed
 - **Deployed Server Type:** The Fep server option is main, backup or None
 - **Simulation:** Enable/Disable data simulation
 - **Add/Remove:** Add/remove server
- **Redundancy:**
 - **Local Management Port:** Declare when the server runs FEP main, specify the redundancy port opened on the local machine.
 - **Remote Management Port:** Declare when the server runs FEP Backup, specify the redundancy port opened at the remote machine.

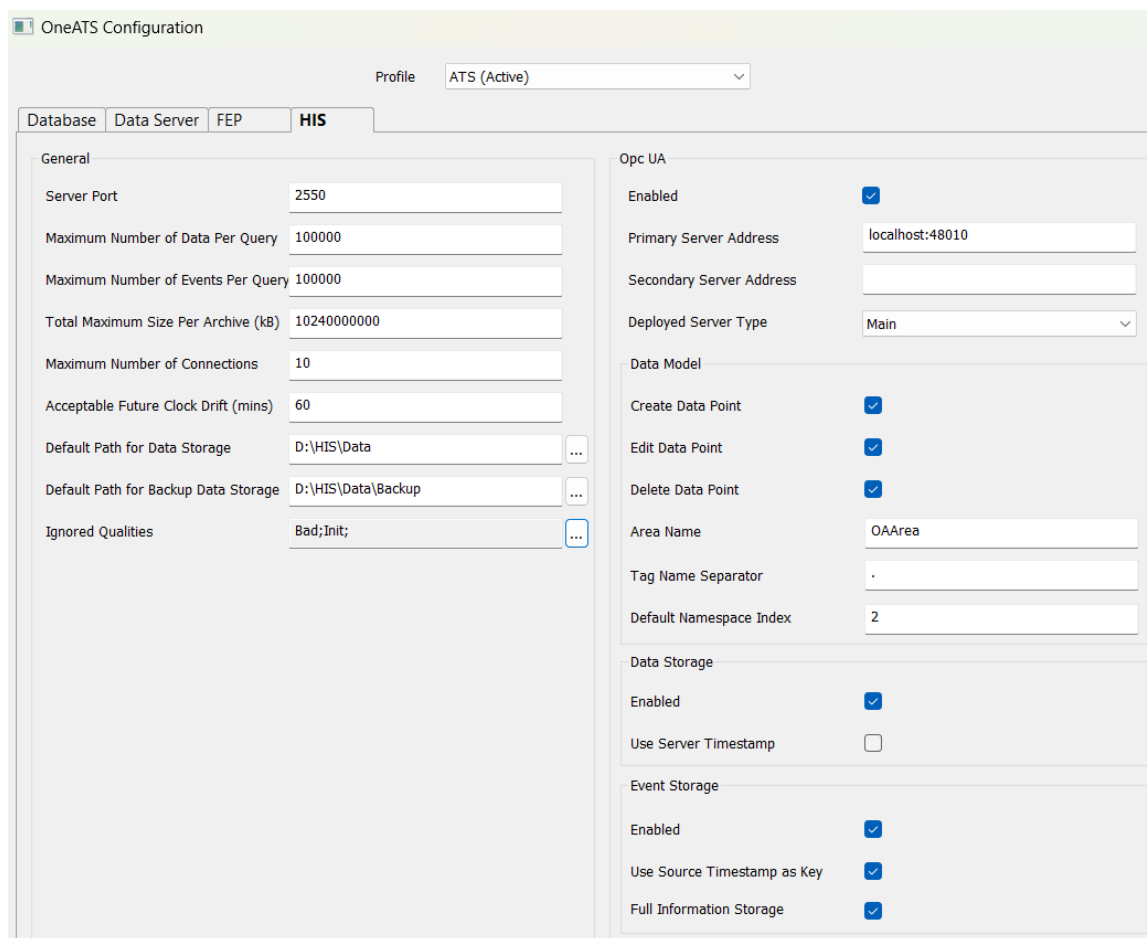
- **Server Timeout (ms):** Control redundancy disconnection time on FEP main side.
- **Client Timeout (ms):** Control redundancy disconnection time on the FEP backup side.
- **Reliable Duration (ms):** Time to evaluate whether the redundancy status is reliable or not.
- **Management:**
 - **Process Management Port:** DE.ProcessingMonitoring data port.
 - **Channel Management Port:** The Fep ProcessManager data port uses the FEP module internally.
 - **Maximum Number of Connections:** Maximum number of connections of DE.ProcessingMonitoring.
- **Management:**
 - **Path:** Path to log file location.
 - **File Size (kB):** Maximum size of 1 log file.
 - **Max Backup Files:** Maximum number of backup log files.
- **Dongle License:** Check/uncheck Dongle license usage.

Select **Apply** to save the configuration file.

The Server parameter configuration matches the **FEP System** configuration on OneATS DataEditor.

5. Configure HIS on OneMasterTool

Run the OneMasterTool application, select **Configuration**, select the **HIS tab**:



The screenshot shows the 'OneATS Configuration' window with the 'HIS' tab selected. The 'Profile' dropdown is set to 'ATS (Active)'. The 'General' section contains fields for 'Server Port' (2550), 'Maximum Number of Data Per Query' (100000), 'Maximum Number of Events Per Query' (100000), 'Total Maximum Size Per Archive (kB)' (10240000000), 'Maximum Number of Connections' (10), 'Acceptable Future Clock Drift (mins)' (60), 'Default Path for Data Storage' (D:\HIS\Data), 'Default Path for Backup Data Storage' (D:\HIS\Data\Backup), and 'Ignored Qualities' (Bad;Init;). The 'Opc UA' section has 'Enabled' checked, 'Primary Server Address' (localhost:48010), 'Secondary Server Address' (empty), 'Deployed Server Type' (Main), and 'Data Model' (Create, Edit, Delete Data Point all checked, Area Name: OAArea, Tag Name Separator: ., Default Namespace Index: 2). The 'Data Storage' section has 'Enabled' checked and 'Use Server Timestamp' unchecked. The 'Event Storage' section has 'Enabled' checked, 'Use Source Timestamp as Key' checked, and 'Full Information Storage' checked.

Figure H-6. HIS Configuration Interface on OneMasterTool

In which:

Parameter	Describe
Server Port	The port on the computer that HIS Server opens the connection to serve
Maximum Number of Data Per Query	Maximum number of data points for a single query
Maximum Number of Events Per Query	Maximum number of events for a single query
Total Maximum Size Per Archive (kB)	Maximum Capacity (KBytes) for all Archives
Maximum Number of Connections	Maximum number of Clients connected to His Server
Acceptable Future Clock Drift (mins)	The period of time ahead of a data point compared to the current time allows the data to still be saved to His Server
Default Path for Data Storage	Where to store His Server data: Log application file, data configuration file, data file

Default Path for Backup Data Storage	Where to store Offline Archives that are automatically backed up	
Ignored Qualities	When querying data from HIS Server, if the data quality is the configured values, it will be ignored or HIS Server will not return that data. In which:	
	Bad	Bad data
	Bad_Override	The overwritten data is of bad quality.
	Bad_Buffer	Bad data written from cache
	Bad_Override_Buffer	Bad quality overwritten data is written from the cache
	Calculate	Calculated data
	Estimate	Estimated data
	Good	Good data
	Good_Override	The overwritten data is of good quality.
	Good_Buffer	Good data is written from the cache
	Good_Override_Buffer	Good quality overwritten data is written from the cache
	Init	Initialization data
	No_Value	Not data
	Overflow	Data exceeds the limit of the data type
	Point_Created	The data of the newly created point
	Shutdown	Data when server is shutdown
Uncertain	Uncertain data	
OPCUa: Configure OPCUa parameters		
Enable	Activate HIS Server's OPCUa ServerConnect to the Data Model (MongoDB Database)	
Primary Server Address	IP/Port of Primary server	
Secondary Server Address	IP/Port of Secondary server	
Deployed Server Type	Server type main/backup/none	
Data Model: Configure Data Model parameters		
Create Data Point	= true : automatically generate points on HIS Server according to the Data Model (MongoDB) = false : do not automatically generate points according to the Data Model (MongoDB). At this time, the configurator manually creates points on the HIS Server according to the needs	
Edit Data Point	= true : allows updating point edits on HIS Server according to Data Model (MongoDB) edits from DataEditor (HIS /Update Data Model function) = false : does not allow updating point edits on HIS Server according to Data Model (MongoDB) edits from DataEditor (HIS /Update Data Model function)	

Delete Data Point	= true: allows deleting a point on the HIS Server when the point is deleted in the Data Model (MongoDB) from the DataEditor (HIS /Update Data Model function) = false: does not allow deleting a point on the HIS Server when the point is deleted in the Data Model (MongoDB) from the DataEditor (HIS /Update Data Model function)
Area Name	The name of the Area is for the data from the Data Model
Tag Name Separator	Character separator of layers in the name of the data point
Default Namespace Index	(OPCUa) NameSpaceIndex of OneATS (Alarm Server) Default value with OneATS system: 2
Data Storage / Enable	= true: allows to receive data (point values) from OPC UA Client (DataServer, ...) = false: does not allow receiving data (point values) from the OPC UA Client (DataServer, ...). (e.g. for Debug operations, only receive Alarm data to check the system)
Data Storage/ Use Server Timestamp	= true: uses the Server Time attribute of the data from the OPC UA Client (DataServer, ...) as the TimeStamp for the data on the HIS Server = false: uses the Device Time attribute of the data from the OPC UA Client (DataServer, ...) as the TimeStamp for the data on the HIS Server
Event Storage / Enable	= true: allows receiving alarm messages from OPC UA Client (DataServer, ...) = false: does not allow receiving alarm messages from the OPC UA Client (DataServer, ...). (e.g. for Debug operations, only receiving data to test the system)
Event Storage / Use Source Timestamp as Key	= true: use the Device Time attribute of the Alarm from the OPC UA Client (DataServer, ...) as the key for the data on the HIS Server to serve the query commands = false: use the Server Time attribute of the Alarm from the OPC UA Client (DataServer, ...) as the key for the data on the HIS Server to serve the query commands
Event Storage / Full Information Storage	= true: saves all NameSpaceIndex of NodeId-type fields (attributes) in Alarm Message = false: does not save NameSpaceIndex of NodeId-type fields (attributes) in Alarm Message. Using DefaultNamespacIndex
Path	Where to save the configuration file

Select **Apply** to save the configuration file.

The Server parameter configuration matches the **HIS System** configuration on OneATS DataEditor.

Area, Account User/Group, Data Set / Digital State, Point, and Archive configurations can be configured to automatically synchronize with Replication mode.

Replication configurations are declared separately for each server

OneATS SmartHIS can operate as a standalone system with *the SmartHIS Administrator configuration tool*

Database, DataServer, FEP, HIS configuration files are stored by default at C:\ProgramData\ATS\OneATS.cfg

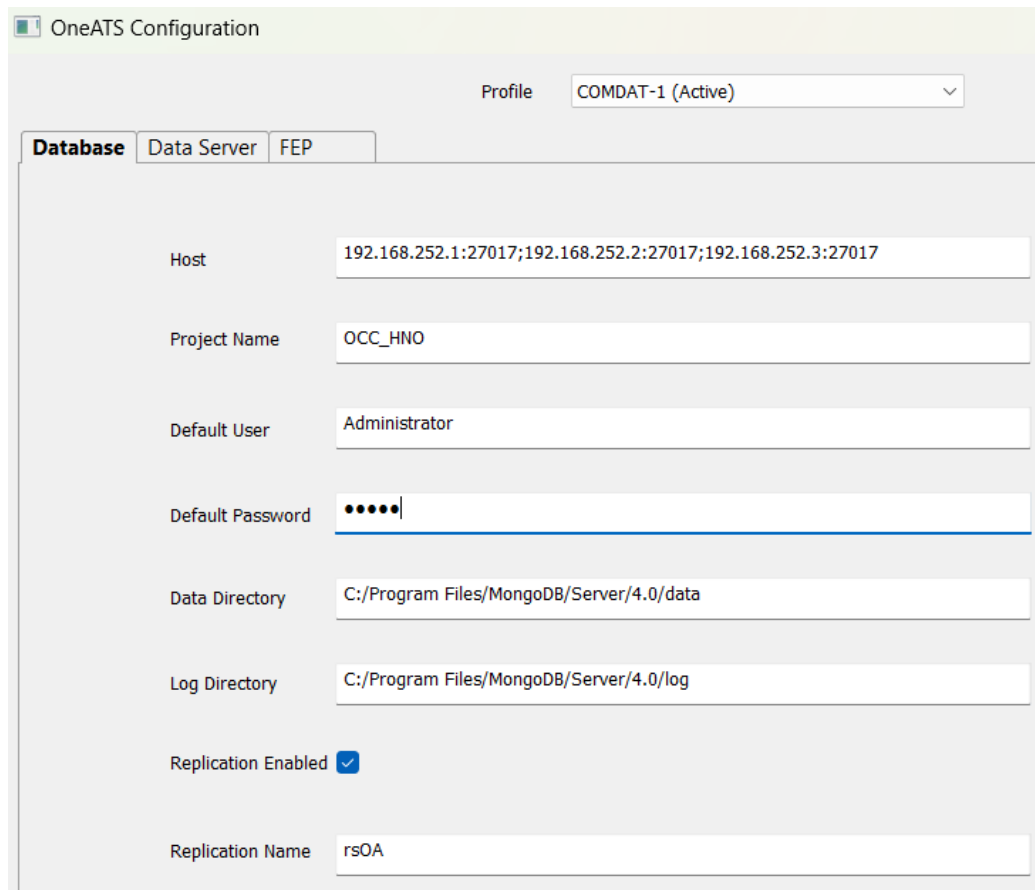
6. Example of deploying software module configuration on OneMasterTool

Example of deploying software module configuration on OneMasterTool with requirements as shown in the following table:

Computer	Description	Installation software	IP Address
COMDAT-1	- Deploy Database on 3 machines COMDAT-1, COMDAT-2, HISAPP-1	Database	192.168.252.1
	- Deploy FEPIO main on COMDAT-1 machine, FEPIO backup on COMDAT-2 machine - Deploy FEPGW1 only on COMDAT-1 machine	FepServer (OneATS FEP)	
	- Deploy SCADA1 main on COMDAT-1 machine, SCADA1 backup on COMDAT-2 machine - Deploy SCADA2 main on COMDAT-1 machine, SCADA1 backup on COMDAT-2 machine - Deploy SCADA3 main on COMDAT-1 machine, SCADA1 backup on COMDAT-2 machine - Deploy SCADA4 main on COMDAT-1 machine, SCADA1 backup on COMDAT-2 machine - Deploy SCADA5 main on COMDAT-2 machine, SCADA1 backup on COMDAT-1 machine - Deploy SCADA6 main on COMDAT-2 machine, SCADA1 backup on COMDAT-1 machine - Deploy SCADA7 main on COMDAT-2 machine, SCADA1 backup on COMDAT-1 machine - Deploy SCADA8 main on COMDAT-2 machine, SCADA1 backup on COMDAT-1 machine	DataServer (OneATS DataServer)	
COMDAT-2	- Deploy Database on 3 machines COMDAT-1, COMDAT-2, HISAPP-1	Database	192.168.252.2
	- Deploy FEPIO main on COMDAT-1 machine, FEPIO backup on COMDAT-1 machine - Deploy FEPGW2 only on COMDAT-2 machine	FepServer (OneATS FEP)	

Computer	Description	Installation software	IP Address
	- Deploy SCADA1 main on COMDAT-1 machine, SCADA1 backup on COMDAT-2 machine - Deploy SCADA2 main on COMDAT-1 machine, SCADA1 backup on COMDAT-2 machine - Deploy SCADA3 main on COMDAT-1 machine, SCADA1 backup on COMDAT-2 machine - Deploy SCADA4 main on COMDAT-1 machine, SCADA1 backup on COMDAT-2 machine - Deploy SCADA5 main on COMDAT-2 machine, SCADA1 backup on COMDAT-1 machine - Deploy SCADA6 main on COMDAT-2 machine, SCADA1 backup on COMDAT-1 machine - Deploy SCADA7 main on COMDAT-2 machine, SCADA1 backup on COMDAT-1 machine - Deploy SCADA8 main on COMDAT-2 machine, SCADA1 backup on COMDAT-1 machine	DataServer (OneATS DataServer)	
HISAPP-1	- Deploy Database on 3 machines COMDAT-1, COMDAT-2, HISAPP-1	Database	192.168.252.3
	- Deploy HIS Server main on HISAPP-1 machine, HIS Server backup on HISAPP-2 machine	OneATS SmartHIS	
HISAPP-2	- Deploy HIS Server main on HISAPP-1 machine, HIS Server backup on HISAPP-2 machine	OneATS SmartHIS	192.168.252.4

- ❖ Configure OneMasterTool on the computer COMDAT-1:
 - Configure Database on OneMasterTool on the computer COMDAT-1:



OneATS Configuration

Profile: COMDAT-1 (Active)

Database | Data Server | FEP

Host: 192.168.252.1:27017;192.168.252.2:27017;192.168.252.3:27017

Project Name: OCC_HNO

Default User: Administrator

Default Password: ●●●●●

Data Directory: C:/Program Files/MongoDB/Server/4.0/data

Log Directory: C:/Program Files/MongoDB/Server/4.0/log

Replication Enabled: ☒

Replication Name: rsOA

Figure H-7. Illustrative interface of Database configuration on COMDAT-1

- Configure DataServer on OneMasterTool on the computer COMDAT-1:

OneATS Configuration

Profile: COMDAT-1 (Active)

Database | **Data Server** | FEP

Server List: 8 Add

Server ID: 1

Server Name: SCADA1

Primary Server Address: 192.168.252.1:48051

Secondary Server Address: 192.168.252.2:48051

Deployed Server Type: Main

Remove

Server ID: 2

Server Name: SCADA2

Primary Server Address: 192.168.252.1:48052

Secondary Server Address: 192.168.252.2:48052

Deployed Server Type: Main

Remove

Server ID: 3

Server Name: SCADA3

Primary Server Address: 192.168.252.1:48053

Secondary Server Address: 192.168.252.2:48053

Deployed Server Type: Main

Remove

Server ID: 4

Redundancy

Mode: Warm

Failover Time (ms): 5000

Auto Revert Enabled: ☒

Alarm & Event

Event Source Format: {PointKey}

Event Message Format: {ParentKey} | {PointDescription}

Operation Log

Path: C:\ProgramData\ATS\OADataServer\log\

File Size (kB): 5120

Max Backup Files: 5

Diagnostic Log

Path: C:\ProgramData\ATS\OADataServer\trace\

File Size (kB): 10240

Max Backup Files: 5

Trace Information	☒	Trace Error	☒
Trace Debug	☐	Trace Data Change	☐
Trace Data Mapping	☐	Trace Data Update	☐
Trace Alarm Update	☐	Trace History Update	☐

OneATS Configuration

ProfileCOMDAT-1 (Active)

DatabaseData ServerFEP

Server List: 8Add

Server ID: 4

Server NameSCADA4

Primary Server Address192.168.252.1:48054

Secondary Server Address192.168.252.2:48054

Deployed Server TypeMain

Remove

Server ID: 5

Server NameSCADA5

Primary Server Address192.168.252.2:48055

Secondary Server Address192.168.252.1:48055

Deployed Server TypeBackup

Remove

Server ID: 6

Server NameSCADA6

Primary Server Address192.168.252.2:48056

Secondary Server Address192.168.252.1:48056

Deployed Server TypeBackup

Remove

Server ID: 7

Redundancy

ModeWarm

Failover Time (ms)5000

Auto Revert Enabled

Alarm & Event

Event Source Format{PointKey}

Event Message Format{ParentKey} | {PointDescription}

Operation Log

PathC:\ProgramData\ATS\ODataServer\log\

File Size (kB)5120

Max Backup Files5

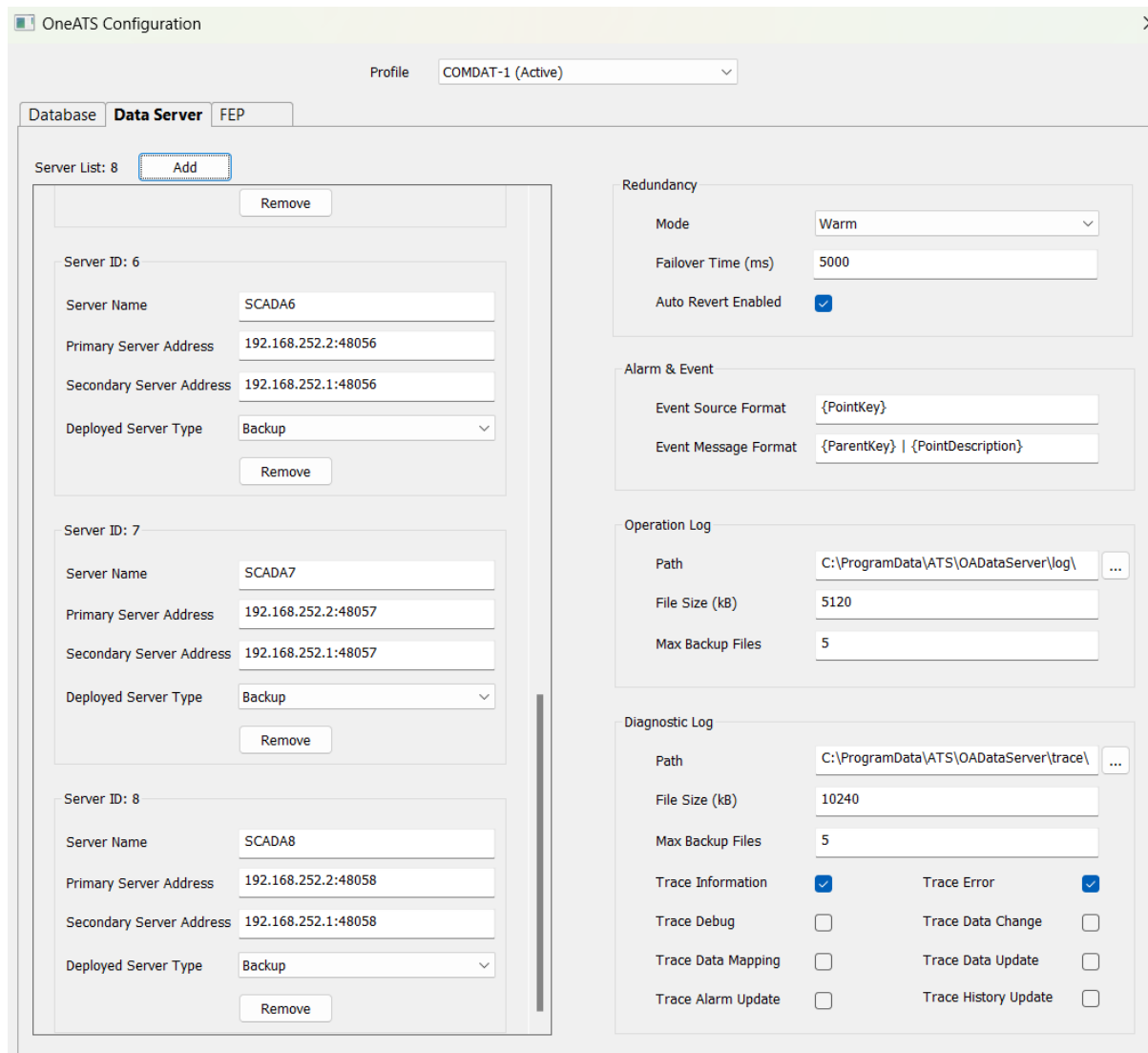
Diagnostic Log

PathC:\ProgramData\ATS\ODataServer\trace\

File Size (kB)10240

Max Backup Files5

Trace InformationTrace ErrorTrace DebugTrace Data MappingTrace Alarm UpdateTrace Data ChangeTrace Data UpdateTrace History Update



OneATS Configuration

Profile: COMDAT-1 (Active)

Database | **Data Server** | FEP

Server List: 8 Add

Remove

Server ID: 6

Server Name: SCADA6

Primary Server Address: 192.168.252.2:48056

Secondary Server Address: 192.168.252.1:48056

Deployed Server Type: Backup

Remove

Server ID: 7

Server Name: SCADA7

Primary Server Address: 192.168.252.2:48057

Secondary Server Address: 192.168.252.1:48057

Deployed Server Type: Backup

Remove

Server ID: 8

Server Name: SCADA8

Primary Server Address: 192.168.252.2:48058

Secondary Server Address: 192.168.252.1:48058

Deployed Server Type: Backup

Remove

Redundancy

Mode: Warm

Failover Time (ms): 5000

Auto Revert Enabled: ☒

Alarm & Event

Event Source Format: {PointKey}

Event Message Format: {ParentKey} | {PointDescription}

Operation Log

Path: C:\ProgramData\ATS\ODataServer\log\

File Size (kB): 5120

Max Backup Files: 5

Diagnostic Log

Path: C:\ProgramData\ATS\ODataServer\trace\

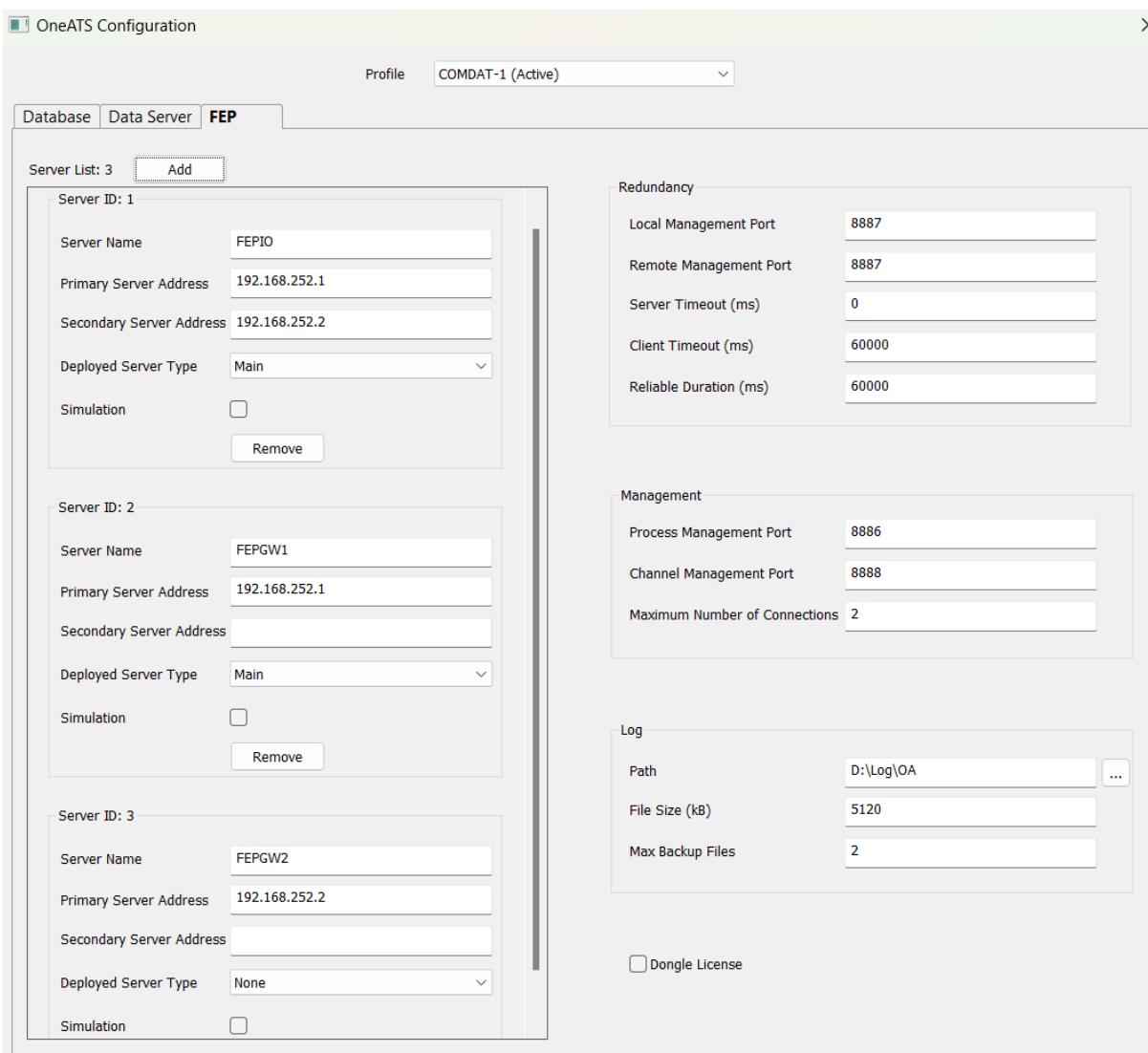
File Size (kB): 10240

Max Backup Files: 5

Trace Information	☒	Trace Error	☒
Trace Debug	☐	Trace Data Change	☐
Trace Data Mapping	☐	Trace Data Update	☐
Trace Alarm Update	☐	Trace History Update	☐

Figure H-8. Illustrative interface of DataServer configuration on COMDAT-1

- Configure FEP on OneMasterTool on the computer COMDAT-1:



OneATS Configuration

Profile: COMDAT-1 (Active)

Database | Data Server | **FEP**

Server List: 3

Server ID: 1

Server Name: FEPIO

Primary Server Address: 192.168.252.1

Secondary Server Address: 192.168.252.2

Deployed Server Type: Main

Simulation: ☐

Server ID: 2

Server Name: FEPGW1

Primary Server Address: 192.168.252.1

Secondary Server Address:

Deployed Server Type: Main

Simulation: ☐

Server ID: 3

Server Name: FEPGW2

Primary Server Address: 192.168.252.2

Secondary Server Address:

Deployed Server Type: None

Simulation: ☐

Redundancy

Local Management Port: 8887

Remote Management Port: 8887

Server Timeout (ms): 0

Client Timeout (ms): 60000

Reliable Duration (ms): 60000

Management

Process Management Port: 8886

Channel Management Port: 8888

Maximum Number of Connections: 2

Log

Path: D:\Log\OA

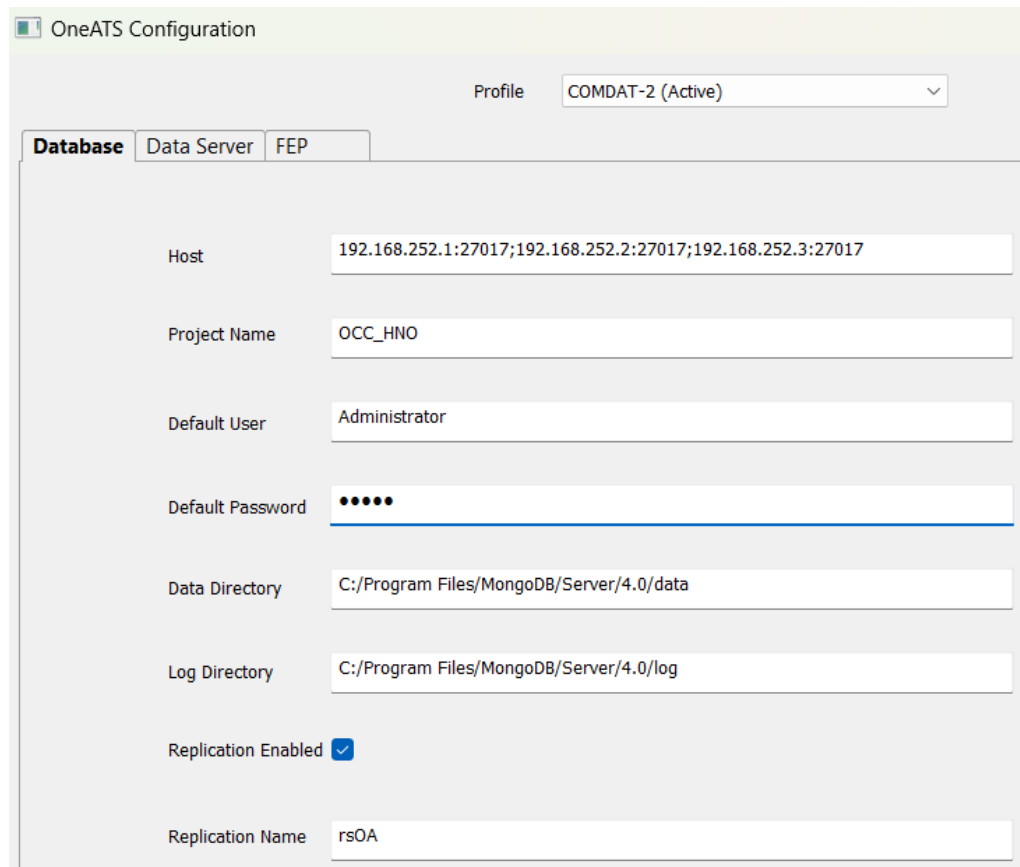
File Size (kB): 5120

Max Backup Files: 2

☐ Dongle License

Figure H-9. Illustrative interface of FEP configuration on COMDAT-1

- ❖ Configure OneMasterTool on the computer COMDAT-2:
 - Configure Database on OneMasterTool on the computer COMDAT-2:



OneATS Configuration

Profile: COMDAT-2 (Active)

Database | Data Server | FEP

Host: 192.168.252.1:27017;192.168.252.2:27017;192.168.252.3:27017

Project Name: OCC_HNO

Default User: Administrator

Default Password: •••••

Data Directory: C:/Program Files/MongoDB/Server/4.0/data

Log Directory: C:/Program Files/MongoDB/Server/4.0/log

Replication Enabled: ☒

Replication Name: rsOA

Figure H-10. Illustrative interface of Database configuration on COMDAT-2

- Configure DataServer on OneMasterTool on the computer COMDAT-2:

OneATS Configuration

ProfileCOMDAT-2 (Active)

DatabaseData ServerFEP

Server List: 8Add

Server ID: 1

Server NameSCADA1

Primary Server Address192.168.252.1:48051

Secondary Server Address192.168.252.2:48051

Deployed Server TypeBackup

Remove

Server ID: 2

Server NameSCADA2

Primary Server Address192.168.252.1:48052

Secondary Server Address192.168.252.2:48052

Deployed Server TypeBackup

Remove

Server ID: 3

Server NameSCADA3

Primary Server Address192.168.252.1:48053

Secondary Server Address192.168.252.2:48053

Deployed Server TypeBackup

Remove

Server ID: 4

Redundancy

ModeWarm

Failover Time (ms)5000

Auto Revert Enabled

Alarm & Event

Event Source Format{PointKey}

Event Message Format{ParentKey} | {PointDescription}

Operation Log

PathC:\ProgramData\ATS\OADataServer\log\

File Size (kB)5120

Max Backup Files5

Diagnostic Log

PathC:\ProgramData\ATS\OADataServer\trace\

File Size (kB)10240

Max Backup Files5

Trace Information

Trace Error

Trace Debug

Trace Data Change

Trace Data Mapping

Trace Data Update

Trace Alarm Update

Trace History Update

OneATS Configuration

ProfileCOMDAT-2 (Active)

DatabaseData ServerFEP

Server List: 8Add

Server ID: 4

Server NameSCADA4

Primary Server Address192.168.252.1:48054

Secondary Server Address192.168.252.2:48054

Deployed Server TypeBackup

Remove

Server ID: 5

Server NameSCADA5

Primary Server Address192.168.252.2:48055

Secondary Server Address192.168.252.1:48055

Deployed Server TypeMain

Remove

Server ID: 6

Server NameSCADA6

Primary Server Address192.168.252.2:48056

Secondary Server Address192.168.252.1:48056

Deployed Server TypeMain

Remove

Server ID: 7

Redundancy

ModeWarm

Failover Time (ms)5000

Auto Revert Enabled

Alarm & Event

Event Source Format{PointKey}

Event Message Format{ParentKey} | {PointDescription}

Operation Log

PathC:\ProgramData\ATS\OADataServer\log\

File Size (kB)5120

Max Backup Files5

Diagnostic Log

PathC:\ProgramData\ATS\OADataServer\trace\

File Size (kB)10240

Max Backup Files5

Trace Information

Trace Error

Trace Debug

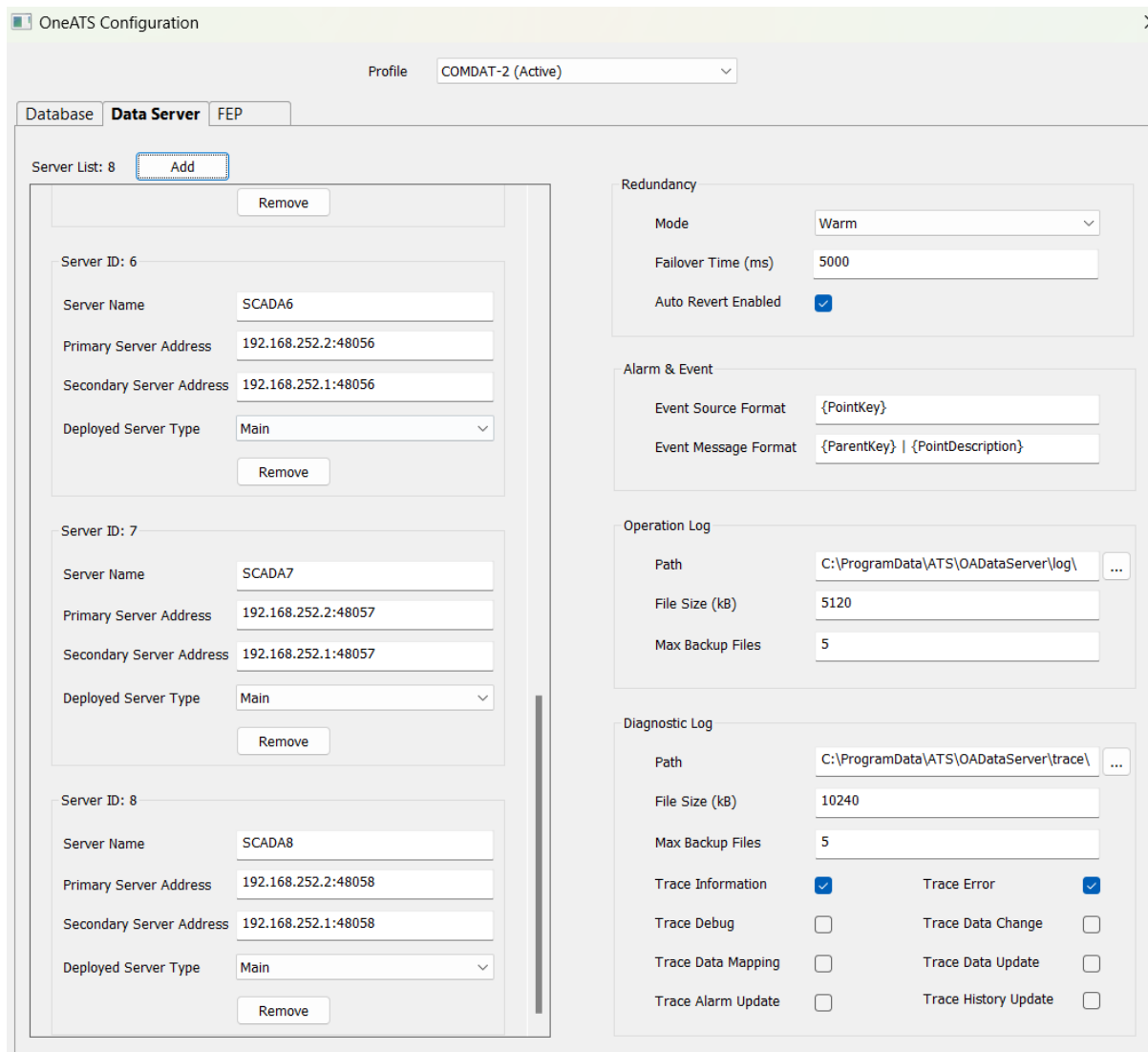
Trace Data Change

Trace Data Mapping

Trace Data Update

Trace Alarm Update

Trace History Update



OneATS Configuration

Profile: COMDAT-2 (Active)

Database | **Data Server** | FEP

Server List: 8 Add

Remove

Server ID: 6

Server Name: SCADA6

Primary Server Address: 192.168.252.2:48056

Secondary Server Address: 192.168.252.1:48056

Deployed Server Type: Main

Remove

Server ID: 7

Server Name: SCADA7

Primary Server Address: 192.168.252.2:48057

Secondary Server Address: 192.168.252.1:48057

Deployed Server Type: Main

Remove

Server ID: 8

Server Name: SCADA8

Primary Server Address: 192.168.252.2:48058

Secondary Server Address: 192.168.252.1:48058

Deployed Server Type: Main

Remove

Redundancy

Mode: Warm

Failover Time (ms): 5000

Auto Revert Enabled: ☒

Alarm & Event

Event Source Format: {PointKey}

Event Message Format: {ParentKey} | {PointDescription}

Operation Log

Path: C:\ProgramData\ATS\ODataServer\log\

File Size (kB): 5120

Max Backup Files: 5

Diagnostic Log

Path: C:\ProgramData\ATS\ODataServer\trace\

File Size (kB): 10240

Max Backup Files: 5

Trace Information: ☒ Trace Error: ☒

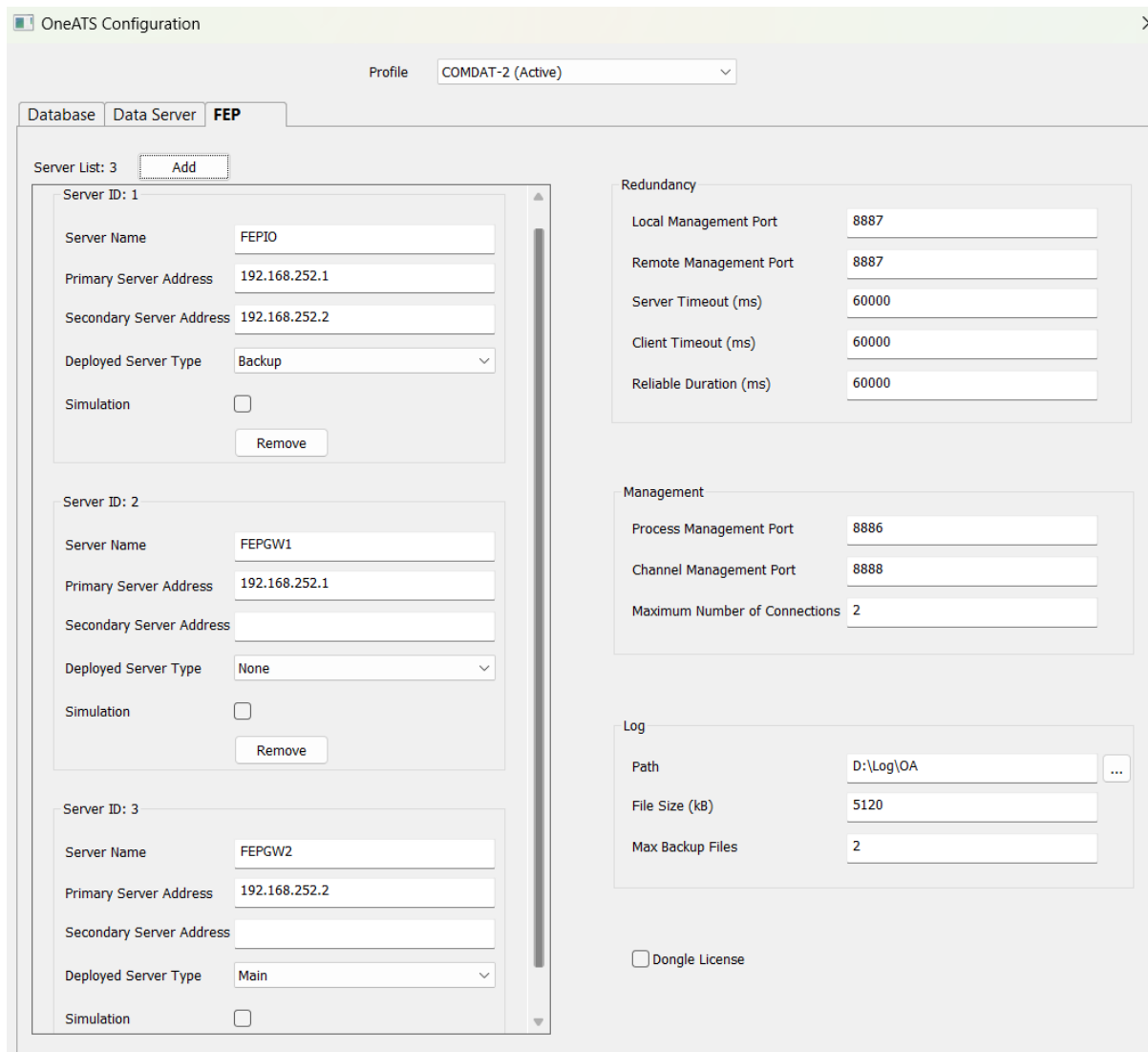
Trace Debug: ☐ Trace Data Change: ☐

Trace Data Mapping: ☐ Trace Data Update: ☐

Trace Alarm Update: ☐ Trace History Update: ☐

Figure H-11. Illustrative interface of DataServer configuration on COMDAT-2

- Configure FEP on OneMasterTool on the computer COMDAT-2:



OneATS Configuration

Profile: COMDAT-2 (Active)

Database | Data Server | **FEP**

Server List: 3

Server ID: 1

Server Name: FEP10

Primary Server Address: 192.168.252.1

Secondary Server Address: 192.168.252.2

Deployed Server Type: Backup

Simulation: ☐

Server ID: 2

Server Name: FEPGW1

Primary Server Address: 192.168.252.1

Secondary Server Address:

Deployed Server Type: None

Simulation: ☐

Server ID: 3

Server Name: FEPGW2

Primary Server Address: 192.168.252.2

Secondary Server Address:

Deployed Server Type: Main

Simulation: ☐

Redundancy

Local Management Port: 8887

Remote Management Port: 8887

Server Timeout (ms): 60000

Client Timeout (ms): 60000

Reliable Duration (ms): 60000

Management

Process Management Port: 8886

Channel Management Port: 8888

Maximum Number of Connections: 2

Log

Path: D:\Log\OA

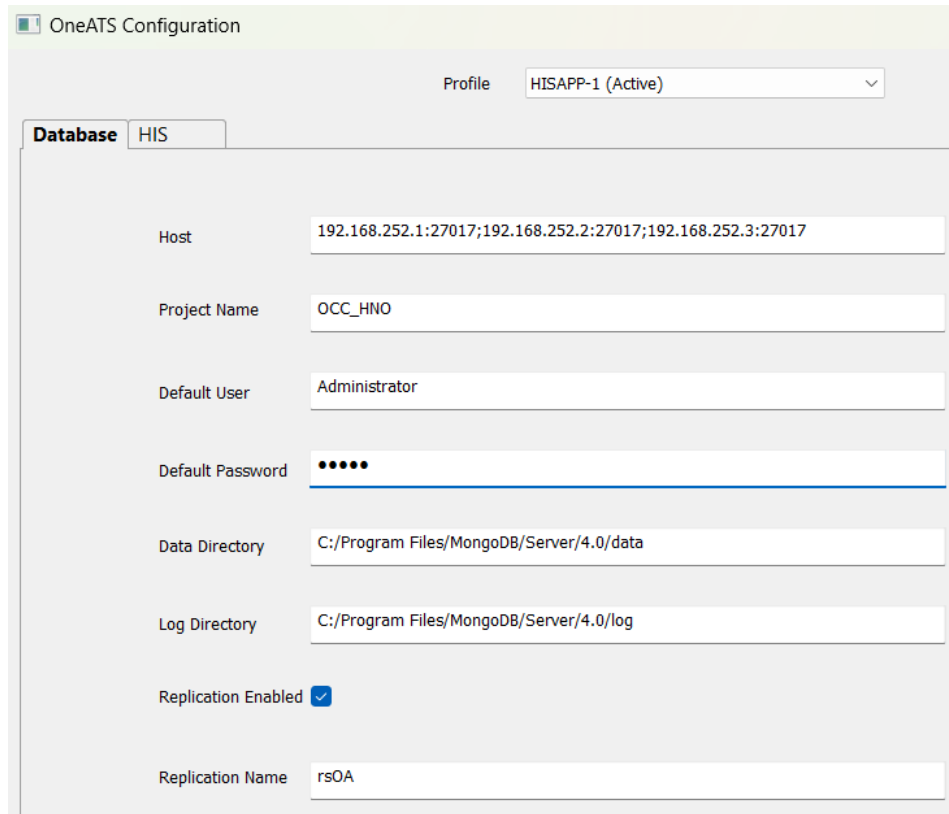
File Size (kB): 5120

Max Backup Files: 2

☐ Dongle License

Figure H-12. Illustrative interface of FEP configuration on COMDAT-2

- ❖ Configure OneMasterTool on the computer HISAPP-1:
 - Configure Database on OneMasterTool on the computer HISAPP-1:

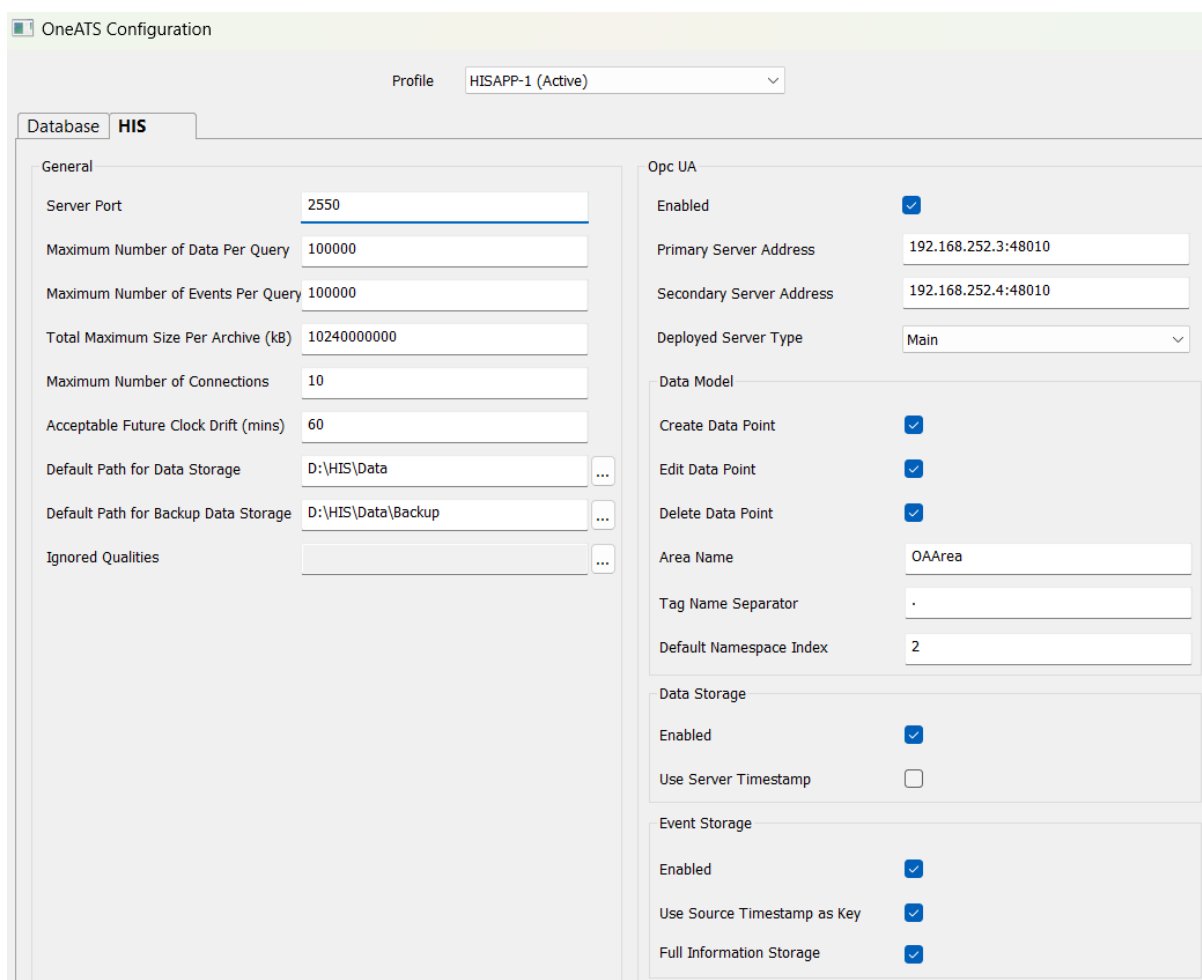


The image shows a screenshot of the 'OneATS Configuration' window. At the top, there is a 'Profile' dropdown menu set to 'HISAPP-1 (Active)'. Below this, there are two tabs: 'Database' and 'HIS'. The 'Database' tab is selected. The configuration fields are as follows:

Field	Value
Host	192.168.252.1:27017;192.168.252.2:27017;192.168.252.3:27017
Project Name	OCC_HNO
Default User	Administrator
Default Password	•••••
Data Directory	C:/Program Files/MongoDB/Server/4.0/data
Log Directory	C:/Program Files/MongoDB/Server/4.0/log
Replication Enabled	☒
Replication Name	rsOA

Figure H-13. Illustrative interface of Database configuration on HISAPP-1

- Configure HIS on OneMasterTool on the computer HISAPP-1:



OneATS Configuration

Profile: HISAPP-1 (Active)

Database: HIS

General

Server Port: 2550

Maximum Number of Data Per Query: 100000

Maximum Number of Events Per Query: 100000

Total Maximum Size Per Archive (kB): 1024000000

Maximum Number of Connections: 10

Acceptable Future Clock Drift (mins): 60

Default Path for Data Storage: D:\HIS\Data

Default Path for Backup Data Storage: D:\HIS\Data\Backup

Ignored Qualities:

Opc UA

Enabled: ☒

Primary Server Address: 192.168.252.3:48010

Secondary Server Address: 192.168.252.4:48010

Deployed Server Type: Main

Data Model

Create Data Point: ☒

Edit Data Point: ☒

Delete Data Point: ☒

Area Name: OAArea

Tag Name Separator: .

Default Namespace Index: 2

Data Storage

Enabled: ☒

Use Server Timestamp: ☐

Event Storage

Enabled: ☒

Use Source Timestamp as Key: ☒

Full Information Storage: ☒

Figure H-14. Illustrative interface of HIS configuration on HISAPP-1

- ❖ Configure OneMasterTool on the computer HISAPP-2:
 - Configure HIS on OneMasterTool on the computer HISAPP-2:

OneATS Configuration

Profile: HISAPP-2 (Active)

Database

HIS

General

Server Port2550

Maximum Number of Data Per Query100000

Maximum Number of Events Per Query100000

Total Maximum Size Per Archive (kB)10240000000

Maximum Number of Connections10

Acceptable Future Clock Drift (mins)60

Default Path for Data StorageD:\HIS\Data

Default Path for Backup Data StorageD:\HIS\Data\Backup

Ignored Qualities

Opc UA

Enabled☒

Primary Server Address192.168.252.3:48010

Secondary Server Address192.168.252.4:48010

Deployed Server TypeBackup

Data Model

Create Data Point☒

Edit Data Point☒

Delete Data Point☒

Area NameOAArea

Tag Name Separator.

Default Namespace Index2

Data Storage

Enabled☒

Use Server Timestamp☐

Event Storage

Enabled☒

Use Source Timestamp as Key☒

Full Information Storage☒

Figure H-15. Illustrative interface of HIS configuration on HISAPP-2

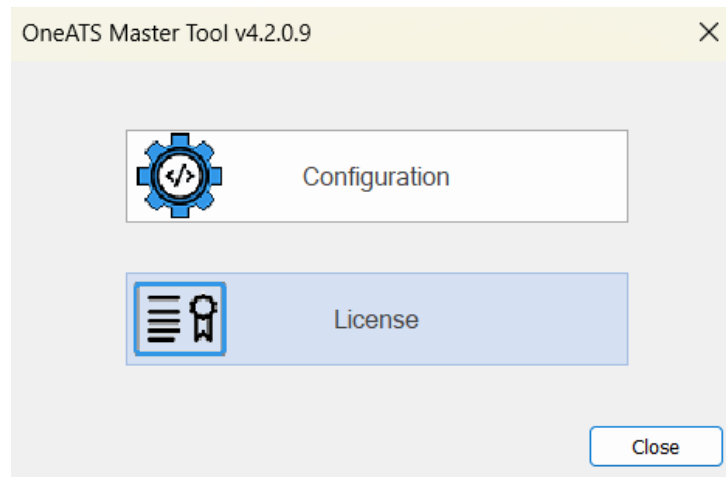
ATS JSC

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IV. Activate the license of the software modules on OneMasterTool

OneMasterTool supports active licenses for software modules in OneATS. Including GRID STUDIO (HMI), DATASERVER, HIS SERVER, FEP SERVER software.

Run the OneMasterTool application, select **License**:



Select **Get**.

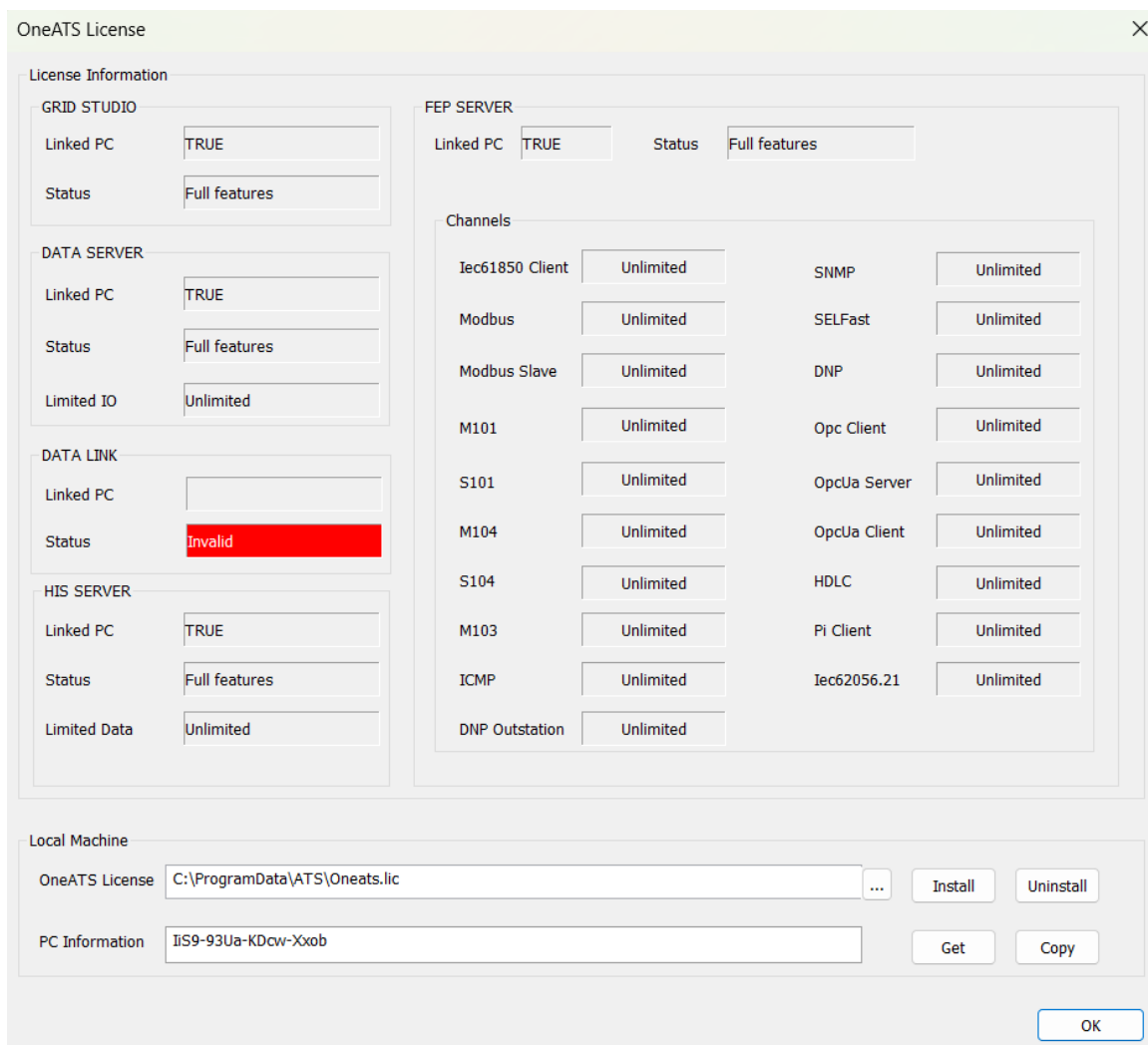
Select **Copy** to copy PC Information

Submit PC Information, project information to ATS for license-registration.

Select the license file from the ...

Select **Install** to install the license.

Select **Uninstal** to clear the existing license.



OneATS License

License Information

GRID STUDIO

Linked PC: TRUE

Status: Full features

DATA SERVER

Linked PC: TRUE

Status: Full features

Limited IO: Unlimited

DATA LINK

Linked PC:

Status: Invalid

HIS SERVER

Linked PC: TRUE

Status: Full features

Limited Data: Unlimited

FEP SERVER

Linked PC: TRUE

Status: Full features

Channels

Iec61850 Client	Unlimited	SNMP	Unlimited
Modbus	Unlimited	SElFast	Unlimited
Modbus Slave	Unlimited	DNP	Unlimited
M101	Unlimited	Opc Client	Unlimited
S101	Unlimited	OpcUa Server	Unlimited
M104	Unlimited	OpcUa Client	Unlimited
S104	Unlimited	HDLC	Unlimited
M103	Unlimited	Pi Client	Unlimited
ICMP	Unlimited	Iec62056.21	Unlimited
DNP Outstation	Unlimited		

Local Machine

OneATS License: C:\ProgramData\ATS\Oneats.lic

PC Information: IIS9-93Ua-KDcw-Xxob

Buttons: Install, Uninstall, Get, Copy, OK

Figure H-16. Active license interface of software on OneMasterTool

Software license information:

GRID STUDIO	
Linked PC	TRUE: the license file matches the computer that is installing the license
Status	- There are 2 statuses: Invalid and Validated. - In case the license has been validated, the Status will display Demo or Full features
DATASERVER	
Linked PC	TRUE: the license file matches the computer that is installing the license
Status	- There are 2 statuses: Invalid and Validated. - In case the license has been validated, the Status will display Demo or Full features
Limited IO	- Unlimited: Unlimited number of IO points- "number": Limit the number of IO points

HIS SERVER	
Linked PC	TRUE: the license file matches the computer on which the license is installed
Status	- There are 2 statuses: Invalid and Validated. - In case the license has been validated, the Status will display Demo or Full features
Limited Data	- Unlimited: Unlimited number of datapoints - "number": Limit the number of datapoints
FEP SERVER	
Linked PC	TRUE: the license file matches the computer on which the license is installed
Status	- There are 2 statuses: Invalid and Validated. - In case the license has been validated, the Status will display Demo or Full features
Unlimited	- Unlimited: Unlimited number of channels/1 protocol - "number": Apply a limit on the number of channels/1 protocol in the Channels section



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